

$$\frac{95}{105} \rightarrow \frac{95}{100}$$

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7-pages

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

(1) (a) Let $\Omega \neq \emptyset$. Define what it means for a collection of subsets of Ω to be:

(i) semi-algebra, (ii) algebra, (iii) σ -algebra, (iv) π -system, (v) λ -system

(b) A nonempty collection $\mathcal{M}$ is said to be a *monotone class* if it is closed under increasing unions and decreasing intersections (i.e. If $\{A_n\}$ is an increasing sequence of sets in $\mathcal{M}$ then $\bigcup_{n \geq 1} A_n \in \mathcal{M}$ and if $\{A_n\}$ is a decreasing sequence of sets in $\mathcal{M}$ then $\bigcap_{n \geq 1} A_n \in \mathcal{M}$). Prove that if $\mathcal{F}$ is a σ -algebra, it is also a

(i) algebra, (ii) π -system, (iii) λ -system, (iv) semi-algebra, (v) monotone class.

(10 × 3 = 30 points)

part 3

part 2

50

65

(2) (a) Let $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$. Define $A_n = \{j : j \geq n\}$, $n \geq 1$, $\mu(A) = |A|$ (counting measure). Show that monotone continuity from above does not hold for this $\{A_n\} \subset \mathcal{F}$.

(b) On the same $(\Omega, \mathcal{F})$ and with the same $\{A_n\} \subset \mathcal{F}$, can you specify a *different* measure μ' such that the monotone continuity from above does hold this $\{A_n\} \subset \mathcal{F}$?

(c) In general, under what conditions does the monotone continuity from above hold?

(d) In general, under what conditions does the monotone continuity from *below* hold?

(4 × 5 = 20 points)

all conditions

(3) (a) Give an example of a function f that is not $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable. — $\mathcal{B}(\mathbb{R})$

(b) Give an example of a function f that is not $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable, but $|f|$ is $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable.

(2 × 5 = 10 points)

(4) (a) Let $\Omega_1 = \mathbb{R}$ and $\mathcal{F}_1 = \{\emptyset, A, A^c, \Omega_1\}$, with $A = (0, 5]$ and let $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Let $T : \Omega_1 \rightarrow \Omega_2$ be defined as

$$T(x) = \begin{cases} 5, & \text{if } x \leq 0 \\ 0, & \text{if } x > 0 \end{cases}$$

Is T $(\mathcal{F}_1, \mathcal{F}_2)$ -measurable? Justify your answer. ΛO

(b) Let $T : \mathbb{R} \rightarrow \mathbb{R}$ be defined as:

$$T(x) = e^{x^2} I_{\{x^2 > 4\}} + \tan(3x) I_{\{\cos(3x) \neq 0\}}.$$

Is T a $(\mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}))$ -measurable transformation? Justify your answer.

yes

$$\lim_{n \rightarrow \infty} \mu(A_n) = \mu(\bigcap_{n \geq 1} A_n)$$

0

$$\mu(A_n) < \infty$$

- ✓ (c) Let each of $X_1, \dots, X_n$ be (real valued) random variables (defined on some $(\Omega, \mathcal{F}, P)$). Then show that $T: \Omega \rightarrow \mathbb{R}$ defined below is a $(\mathcal{F}, \mathcal{B}(\mathbb{R}))$ -measurable transformation:

$$T(\omega) = \frac{\sqrt{\frac{1}{n-1} \left(\sum_{i=1}^n X_i^2(\omega) - \left(\frac{\sum_{i=1}^n X_i(\omega)}{n} \right)^2 \right)}}{\left(\frac{1}{n} \sum_{i=1}^n X_i(\omega) \right)} \mathbf{I}_{\left\{ \frac{1}{n} \sum_{i=1}^n X_i(\omega) \neq 0 \right\}}$$

sample variance

[Note: in statistics, this is the standard "plug-in estimate" for the "coefficient of variation" $= \sigma/\mu$ of a random variable, where μ, σ are mean and s.d of the random variable]
(3 × 10 = 30 points)

- (5) Take $F(x) = (1 - e^{-x}) \mathbf{I}_{\{x \geq 0\}}$.

how to
get
extension (a) Using the steps in Caratheodory Extension Theorem(s), show by starting from defining a measure μ_F on a suitable semi-algebra, how one can extend it to the Lebesgue-Stieltjes measure μ_F^* on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

- (b) Is such a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ unique? Why?

[Write down the steps of the extension theorem carefully and state the results used in each step]

(10 + 5 = 15 points)

add later